Prob. 4: Consider the circuit shown where $V_S = 20V$ and $I_S = 4A$. Using *mesh method*.

a. Write all the required equations to find i_1, i_2, i_3, and v_o (*Do not simplify*)

b. Solve to find i_1, i_2, i_3, and v_o.

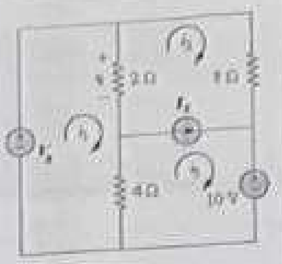

Figure 4

a.

$8I_2 + 10 + 4(I_3 - I_1) - 2(I_2 - I_1)$

$-2I_1 + 6I_2 + 4I_3 = -10 \Rightarrow (1)$

$I_3 + I_S = I_2$

$I_3 + 4 = I_2$

$I_2 - I_3 = 4 \Rightarrow (2)$

$2(I_1 - I_2) + 4(I_1 - I_3) = -V_S$

$6I_1 - 2I_2 - 4I_3 = -20 \Rightarrow (3)$

b.

$i_1 =$ -6.75 A ; $i_2 =$ -0.75 A

$i_3 =$ -4.75 A ; $v_o =$ $2(I_1 - I_2) =$ -12V

6

Prob. 1: Consider the circuit shown where V_S = 30V.

a. Using voltage divider, find V_o and V_{ab}.

b. Using Ohm's law, find I_S and I.

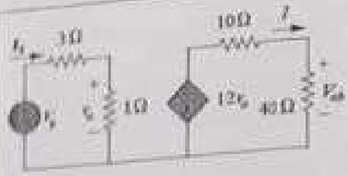

Figure 1

a.

V_o = 7.5V

V_{ab} = -72 V

6

b.

$I_S = \frac{V_s}{R_{eq}} = \frac{30}{4} = 7.5A$

$I = \frac{-12V_o}{R_{eq}} = \frac{-90}{50} = -1.8A$

Prob. 2: Consider the circuit shown where R = 14 Ω.

a. Find the equivalent resistance between the a-b terminals.

b. When the voltage between the *a-b* terminals is 84 V, find the current I and the voltage V_x.

c. When I = 10A, using current division, find the current through the 30-Ω resistance.

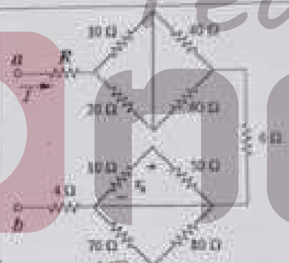

Figure 2

a.

R_{eq} = 61.33Ω

b.

$I = \frac{V}{R_{eq}} = \frac{84}{61.33} = 1.367A$

V_x = 0V (short circuit)

c.

$I_{30} = \frac{V}{R_{30}} = \frac{84}{30} = 2.8A$

Prob.3:

(a) The charge transferred through a circuit element is givin by $q(t)=6t^2+6t$ C Calculate the current entering the element at instant $t = 10ms$.

$i(t)$ = 126 A

(b) If a load draws 1kW when connected to a 230 V source, then the resistance of the load (R_L) is

R_L = 15.16Ω

Prob. 5: Consider the circuit shown where $I_s = 2A$. Using **nodal method**:

a. Write **all** the required equations to find v_1, v_2, v_3, and I_o (*Do not simplify*)

b. Solve to find v_1, v_2, v_3, and I_o.

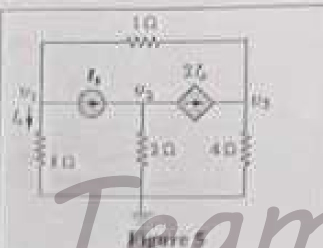

Figure 5

a.

$$I_o = \frac{V_1}{8}$$

$$I_o + I_s + \frac{V_1 - V_3}{1} = 0$$

$$I_o + V_1 - V_3 = -2$$

$$\frac{7V_1}{8} - V_3 = -2 \Rightarrow (1)$$

$$-I_s + \frac{V_2}{2} + 2I_o = 0$$

$$2I_o + \frac{V_2}{2} = 2$$

$$\frac{7V_1}{4} + \frac{V_2}{2} = 2 \Rightarrow (2)$$

$$\frac{V_3}{4} + \frac{V_3 - V_1}{1} - 2I_o = 0$$

$$-V_1 + \frac{5V_3}{4} - 2I_o = 0$$

$$\frac{-11V_1}{4} + \frac{5V_3}{4} = 0 \Rightarrow (3)$$

b.

$v_1 = 1.5\ V$; $v_2 = -1.28V$

$v_3 = 3.32V$; $I_o = \frac{1.5}{8} = 0.1875A$